1 Results

1.1 Gathered data

- Number of data available for angiosperm and gymnosperm originally from Silvics book, with masting data, and the structure of data
  - Angiosperm (In the book: 124; with masting data: 98; Masting/Non-masting: (51/47); Gymnosperm (In the book: 64; with masting data: 61; Masting/Non-masting: (57/7))
  - Strong masting more common for gymnosperm
- Number of traits we have for different hypotheses for angiosperm and gymnosperm
  - For angiosperm, we have data for:
    * Predation satiation: seed weight, fruit size, oil content, seed dispersal mode, seed dormancy
    * Pollination coupling: pollination mode, reproductive type
    * Resource matching: drought tolerance, leaf longevity
  - For gymnosperm, we only have oil content for masting species and all the gymnosperm species are wind pollinated.

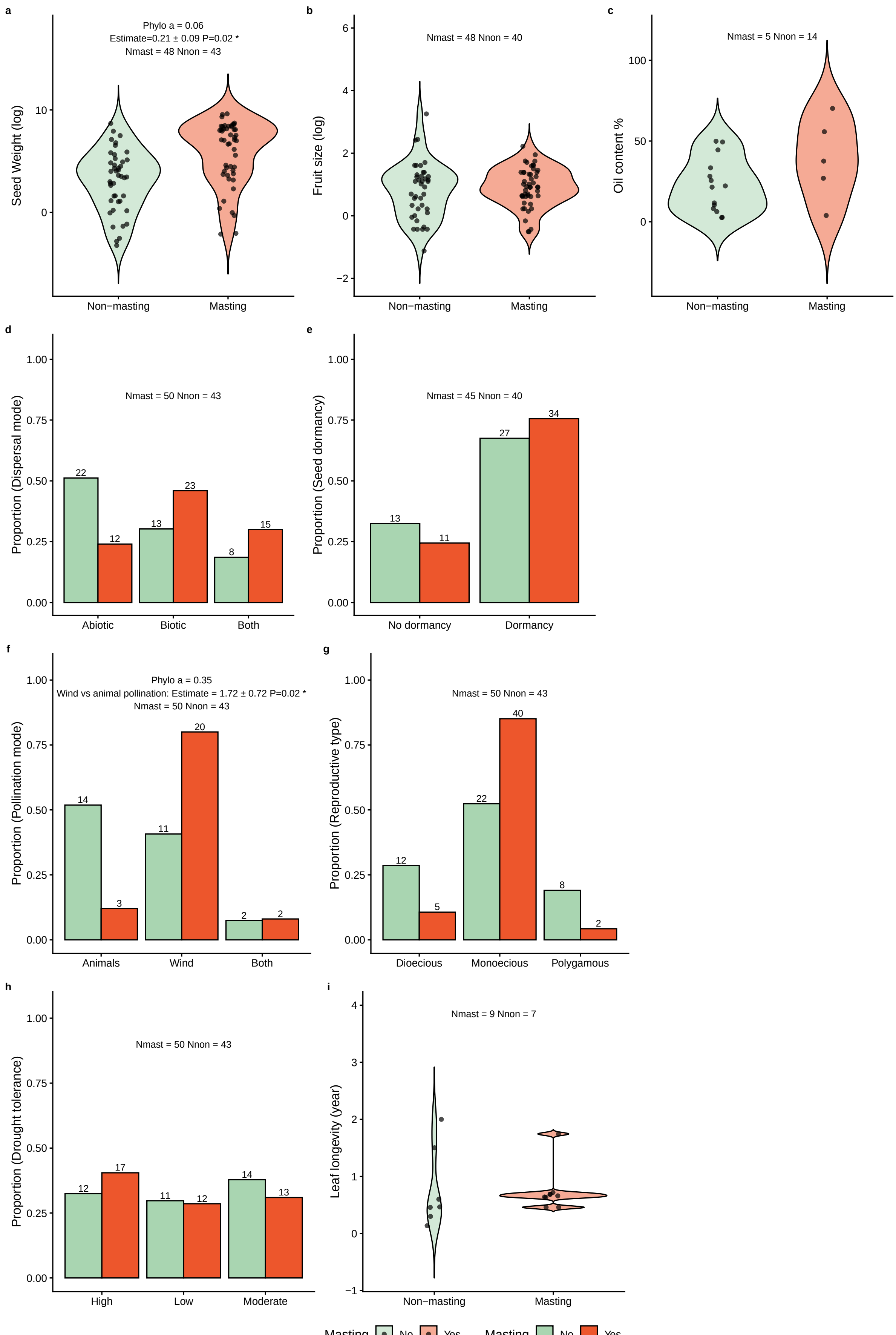

Figure 1: Relationships between plant traits (a) Seed weight; (b) Fruit size; (c) Oil content; (d) Dispersal mode; (e) Seed dormancy; (f) Pollination mode; (g) Reproductive mode; (h) Drought tolerance; (i) Leaf

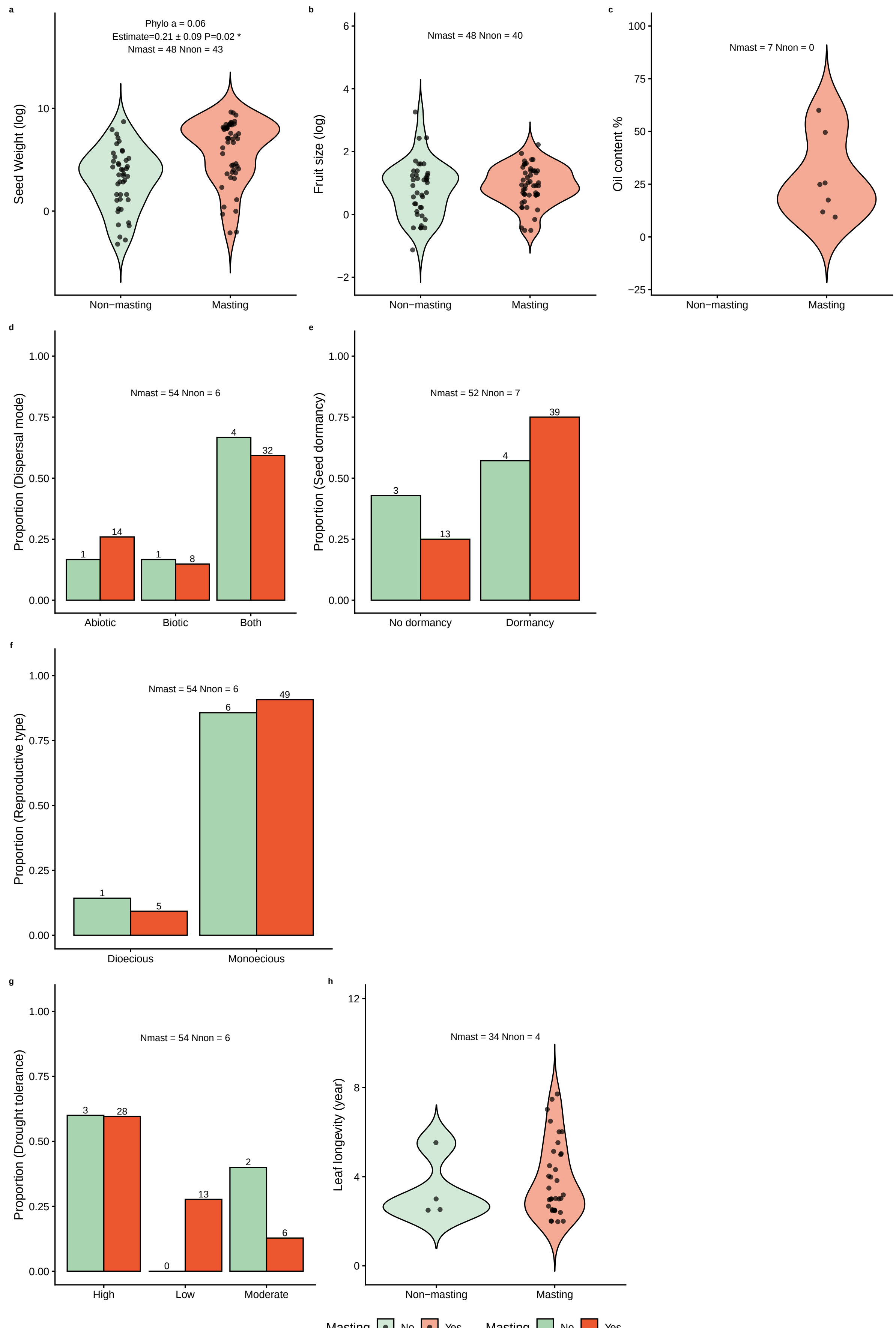

Figure 2: Relationships between plant traits (a) Seed weight; (b) Fruit size; (c) Oil content; (d) Dispersal mode; (e) Seed dormancy; (f) Pollination mode; (g) Reproductive mode; (h) Drought tolerance; (i) Leaf

Trait	Predictor	Estimate	SE	P	Phylo α	N
Seed dispersal	Biotic vs abiotic	-0.83	0.81	0.30	0.01	60
Seed dispersal	Both vs abiotic	-1.11	0.67	0.10	0.01	60
Seed dormancy	Dormant vs non-dormant	0.71	0.70	0.32	0.41	59
Reproductive type	Monoecious vs dioecious	-1.14	1.02	0.26	0.01	61
Seed weight (log)	logSeedWeight	-0.03	0.25	0.91	0.43	59
Fruit size (log)	logFruit	0.03	0.36	0.94	0.43	60
Seed size (log)	logSeedSize	-0.24	0.92	0.80	0.41	51
Leaf longevity (years)	Leaf longevity (years)	0.15	0.34	0.67	0.43	38
Drought tolerance	Low vs high drought tolerance	0.91	1.64	0.58	0.43	52
Drought tolerance	Moderate vs high drought tolerance	-1.56	1.06	0.14	0.43	52

Table 1: Phylogenetic Generalized Linear Model Results for Gymnosperm

Trait	Predictor	Estimate	SE	P	Phylo α	N
Dispersal mode	Biotic vs abiotic	0.55	0.53	0.30	0.06	93
Dispersal mode	Both vs abiotic	0.55	0.57	0.33	0.06	93
Pollination mode	Wind vs animal pollination	1.72	0.72	0.02	0.35	52
Pollination mode	Wind vs animal pollination and animals	1.23	1.15	0.29	0.35	52
Seed dormancy	Dormant vs non-dormant	0.03	0.50	0.95	0.10	85
Reproductive type	Monoecious vs dioecious	0.90	0.68	0.18	0.11	89
Reproductive type	Monoecious vs polygamous	-0.13	0.96	0.89	0.11	89
Seed weight (log)	logSeedWeight	0.21	0.09	0.02	0.06	91
Fruit size (log)	logFruit	0.08	0.30	0.80	0.17	88
Seed size (log)	logSeedSize	0.24	0.30	0.43	0.08	69
Oil content (%)	Oil content (%)	0.04	0.03	0.20	0.14	19
Leaf longevity (years)	Leaf longevity (years)	0.26	0.95	0.78	0.01	16
Drought tolerance	Low vs high drought tolerance	0.04	0.48	0.92	0.11	79
Drought tolerance	Moderate vs high drought tolerance	-0.13	0.46	0.79	0.11	79

Table 2: Phylogenetic Generalized Linear Model Results for Angiosperm

1.2 Results for each hypothesis

- Predation satiation: seed weight only with a small log-odds and only for angiosperm
  - We found only seed weight is related to masting with having bigger seeds will increase the chance of being a strong masting species Figure (1), however, the log-odds is only 0.2 Table (**??**). We also only found this in angiosperm group.
- Pollination satiation: wind pollination for angiosperm
  - we found wind pollination is related to masting for angiosperm species, the log-odds for wind compare to animal or both are 1.72 and 1.23 respectively. Gymnosperm are all wind pollinated, and gymnosperm has a greater portion of masting species.
- Resource matching: nothing significant

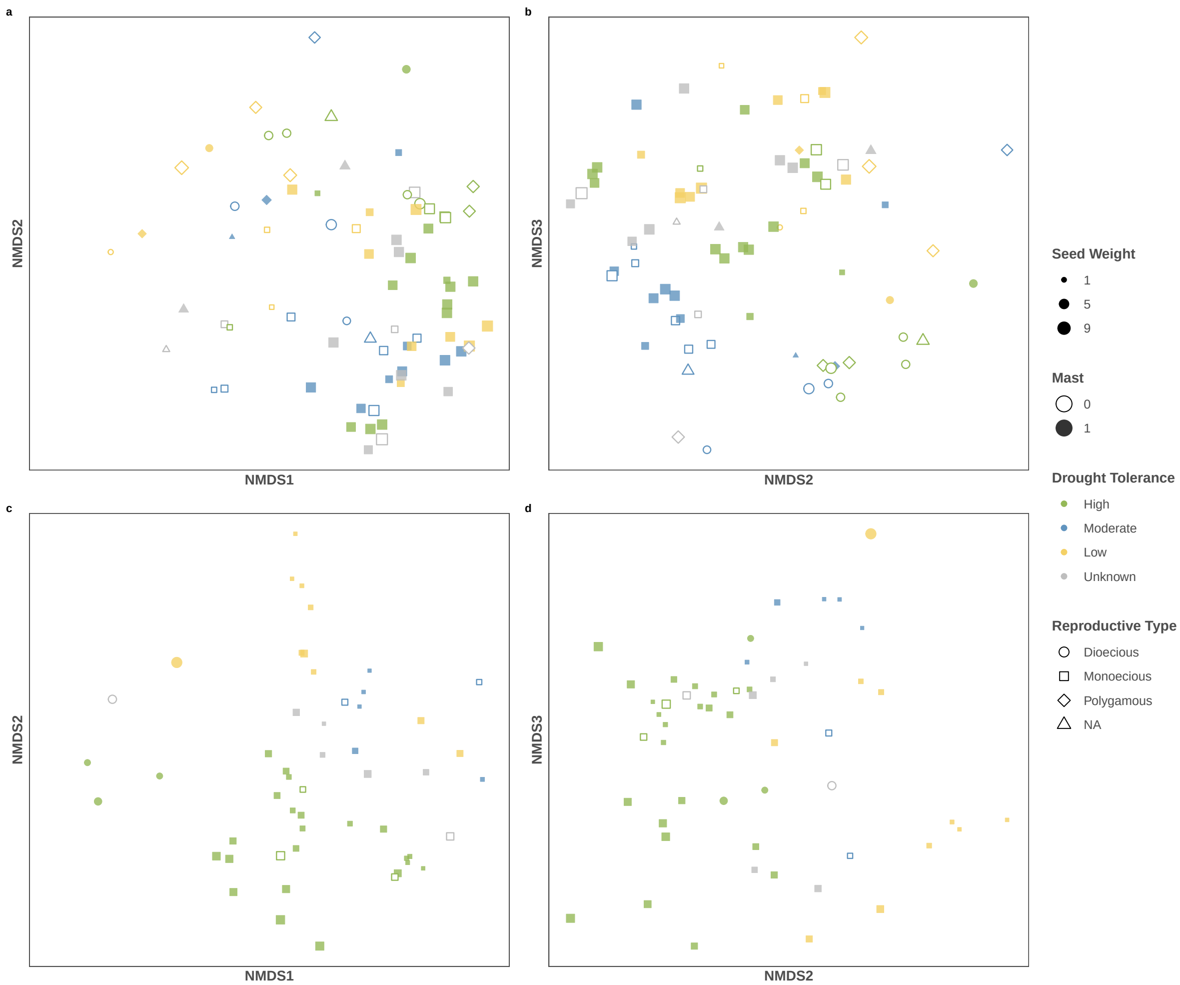

Figure 3: Non-metric multidimensional scaling (NMDS) of trait composition for angiosperm and conifer species. Each point represents a species, with position determined by NMDS axes. Points are colored by drought tolerance, filled by masting or not, shaped by reproductive type, sized by seed weight. Panels are arranged as (a) angiosperm NMDS1–NMDS2, (b) angiosperm NMDS2–NMDS3, (c) conifer NMDS1–NMDS2, and (d) conifer NMDS2–NMDS3.

1.3 Multidimension-NMDS

- We don't see masting aligned well along any axis in the NMDS analysis.

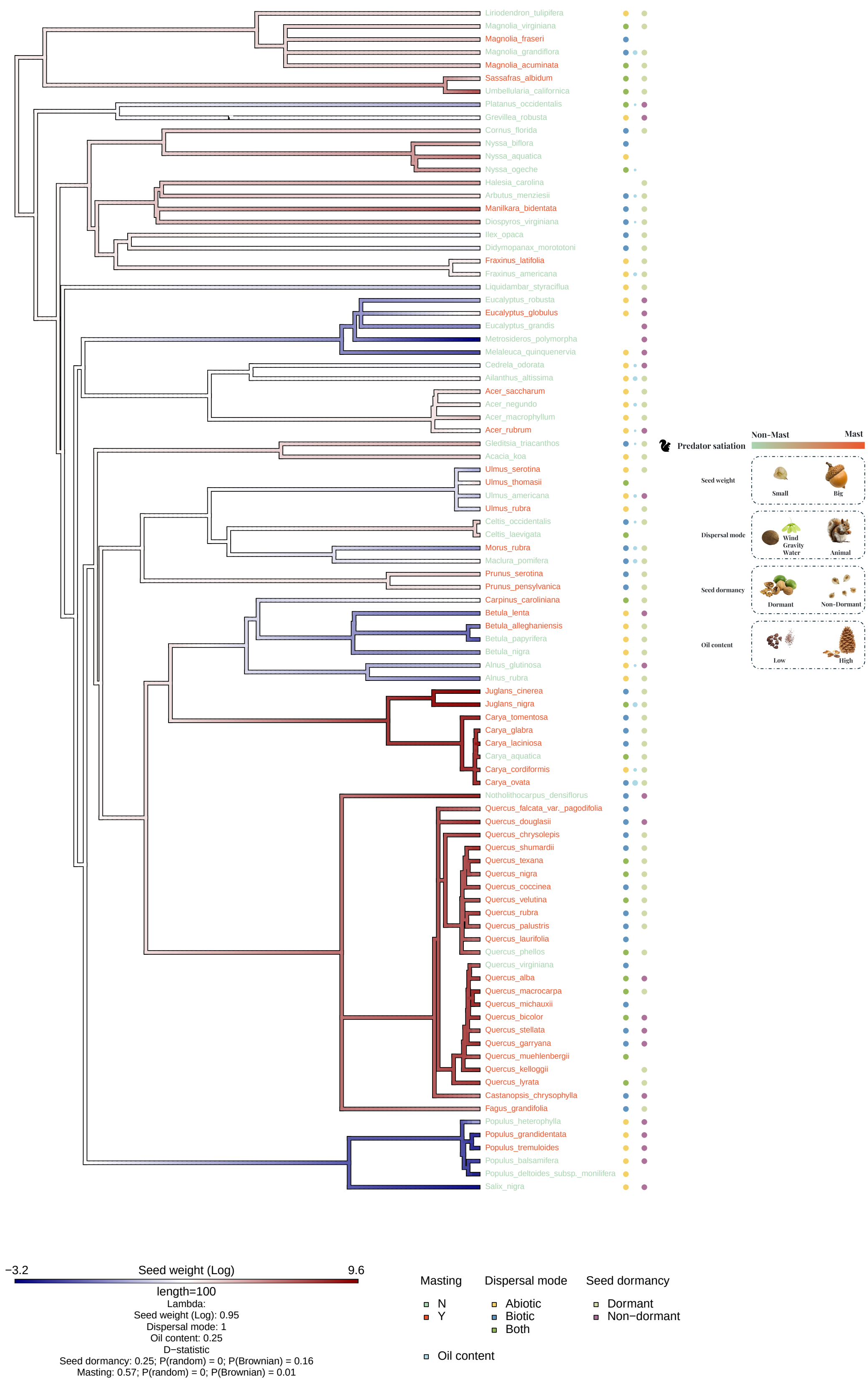

Figure 4: Phylogenetic tree for all species included in the analysis. The color scheme of the branch is the trait value for seed weight (log). The color tip is colored by masting or not. The dispersal mode, oil content and seed dormancy are labeled after the tip labels with different color and size. We also show the predictions of relationship between traits and masting. We calculated the Pagel's λ for continuous traits as well as multilevel categorical traits, and calculated D-statistic for binary traits

Figure 5: Phylogenetic tree for all Quercus species, Quercus species included in this study is color coded.

Group	Trait	(λ)
Angiosperm	Seed weight (Log)	0.95
Angiosperm	Fruit size (Log)	0.93
Angiosperm	Oil content	0.25
Angiosperm	Leaf longevity	1.02
Angiosperm	Dispersal mode	1.00
Angiosperm	Pollination mode	1.00
Angiosperm	Reproductive type	1.00
Angiosperm	Drought tolerance	0.00
Gymnosperm	Seed weight (Log)	0.97
Gymnosperm	Fruit size (Log)	0.90
Gymnosperm	Oil content	0.00
Gymnosperm	Leaf longevity	0.90
Gymnosperm	Dispersal mode	1.00
Gymnosperm	Drought tolerance	1.00

Table 3: Phylogenetic Signal (Pagel's λ) for continous traits and multi-level categorical traits

Group	Trait	Estimated D	P(random)	P(Brownian)
Angiosperm	Seed dormancy	0.25	0	0.16
Angiosperm	Masting	0.58	0	0.01
Gymnosperm	Masting	0.51	0.02	0.10
Gymnosperm	Seed dormancy	0.98	0.41	0
Gymnosperm	Reproductive type	-0.52	0	0.89

Table 4: Phylogenetic Signal (D-statistic with probability from random phylogenetic structure and probability from Brownian phylogenetic structure for binary traits

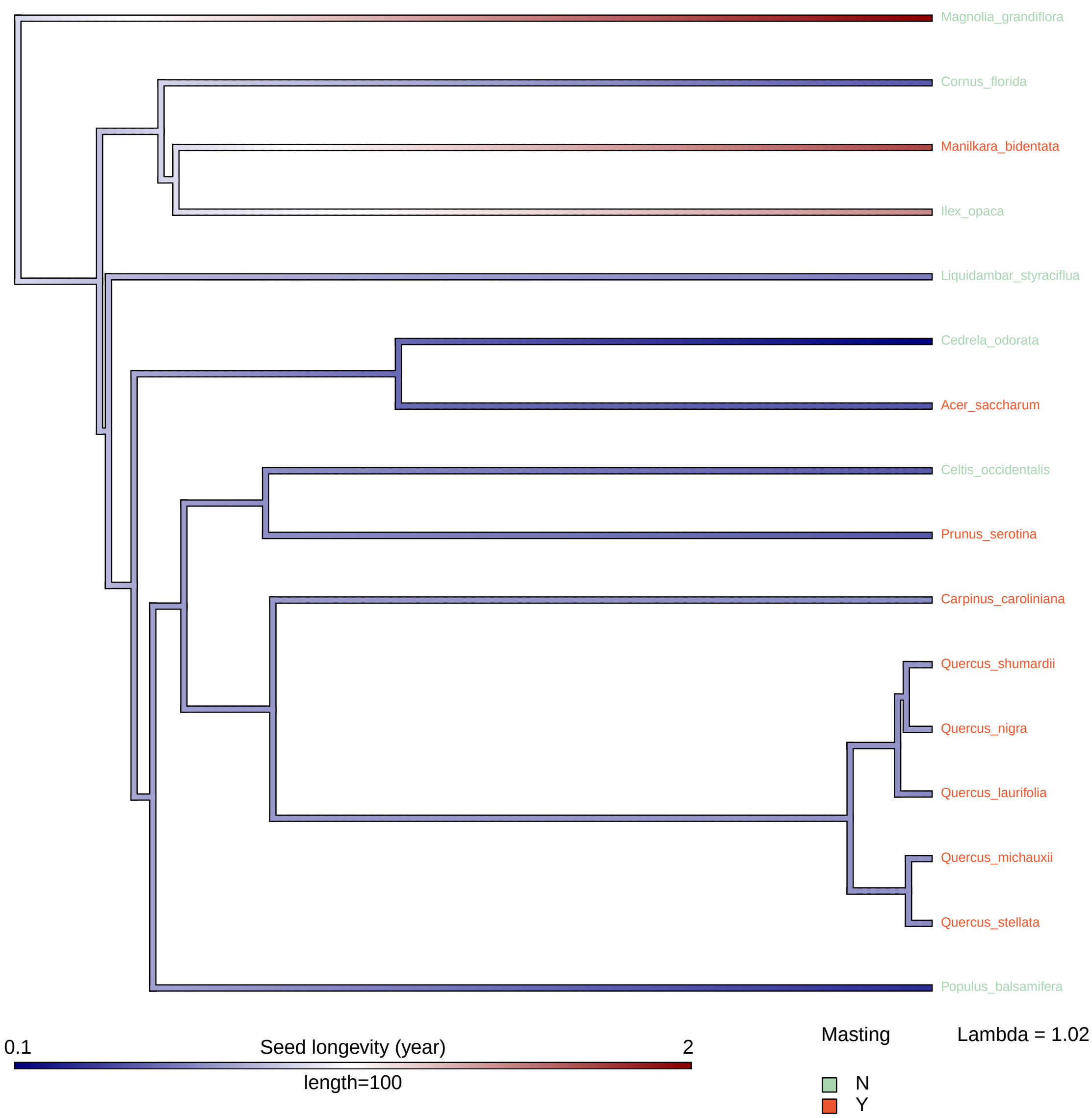

Figure 6: Phylogenetic tree for species with leaf longevity data, tip label color-coded by masting pattern

1.4 Phylogenetic patterns

- Pagel's λ for continuous traits and multilevel categorical traits in Table 3, D-statistic for binary traits in Table 4.
  - Masting has a stronger phylogenetic signal in gymnosperm compared to angiosperm
  - Seed weight has strong phylogenetic signal and we can see the similarity in certain group in Figure 4 it's relatively reliable estimate
  - Be cautious that λ might not be comparable among traits, not only because the way λ is calculated for continuous traits and categorical traits are different, but also because λ might depend on the data structure when sampling is uneven, take leaf longevity as an example in Figure 6.
- When we know the phylogenetic results, we need to be cautious interpretating relationships between traits, it is hard to tease out traits from masting, limit the inference)
- Data is strongly nested in phylogeny, and data is sparse. We have to assume the sampling is even.